Quiz 2: Algorithm Analysis & Recurrences

SCS1308 Foundations of Algorithms

Question 1

Which of the following recurrences exhibit exponential time growth?

a. $T(n) = T(n/2) + T(n/3) + O(1)$

b. **$T(n) = 3T(n/2) + O(1)$** CORRECT

c. **$T(n) = 2T(n-1) + O(1)$** CORRECT

d. $T(n) = T(n-1) + O(1)$

e. $T(n) = T(n/2) + O(1)$

Question 2

Which of the following are valid base cases for recurrences?

a. **$T(n) = 0$ for $n = 1$** CORRECT

b. **$T(0) = O(1)$** CORRECT

c. **$T(1) = O(1)$** CORRECT

d. **$T(n) = O(1)$ for all $n < c$** CORRECT

e. $T(n) = T(n-1)$

Question 3

The recurrence $T(n) = T(n-1) + O(1)$ corresponds to:

a. $O(n!)$

b. $O(n^2)$

c. $O(2^n)$

d. **$O(n)$** CORRECT

e. $O(\log n)$

Question 4

Which of the following statements are TRUE for Big-O notation?

a. **It can hide constant factors.** CORRECT

b. **It gives an asymptotic upper bound.** CORRECT

c. It cannot be used for recursive functions

Question 6

Which of the following recurrences solves to Θ(n log n)?

a. **T(n) = 2T(n/2) + n** CORRECT

b. T(n) = 2T(n/2) + n^2

c. T(n) = T(n/2) + n

d. T(n) = T(n−1) + O(1)

e. T(n) = 3T(n/2) + n

Question 7

If T(n) = T(n/2) + O(1), what is the asymptotic complexity?

a. **O(log n)** CORRECT

b. O(n log n)

c. O(√n)

d. O(n)

e. O(1)

Question 8

Consider T(n) = T(n/2) + T(n/4) + O(n). What is the complexity?

a. **O(n)** CORRECT

b. O(n^2)

c. O(n log n)

d. O(log n)

e. O(n^3)

Question 9

For the recurrence T(n) = 2T(n/2) + √n, the complexity is:

a. O(√n log n)

b. **O(n)** CORRECT

c. O(n√n)

d. O(n log n)

e. O(log n)